



MANIPAL ACADEMY OF HIGHER EDUCATION

Manipal Institute of Technology, Bengaluru

BTech VI Semester

End Semester Examination - April / May -2025

APPLICATIONS OF AI IN MEDICINE [CSE- 4014]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A machine learning model is used to classify chemical compounds as either cancer-killing or not cancer-killing. Based on its predictions: It correctly identified 85 actual cancer-killing compounds. It mistakenly labelled 15 non-cancer-killing compounds as cancer-killing. It failed to detect 10 cancer-killing compounds, predicting them as non-cancer-killing.
A) It correctly labelled 90 non-cancer-killing compounds.
Considering that missing a cancer-killing compound (false negative) could result in a lost opportunity for a potential treatment, while misclassifying a non-cancer-killing compound as effective (false positive) may just lead to wasted lab resources: (4)
1. Calculate the precision, recall, F1-score and accuracy for the cancer-killing class.
 2. Analyse whether the model's recall is sufficient given the higher cost of false negatives.
 3. Suggest two techniques to improve recall, and explain the possible trade-offs involved.
- B) A pharmaceutical research team is using machine learning to identify potential cancer-killing compounds from a large dataset of chemical structures. The dataset is labelled based on prior laboratory testing and includes both cancer-killing and non-cancer-killing compounds. To automate classification of new compounds, the team chooses to implement a Support Vector Machine (SVM) classifier. As a data scientist on the team, provide a comprehensive explanation of how the SVM algorithm would be applied in this context along with mathematical and geometric intuition behind how SVM separates the two classes and the role of support vectors, maximum margin hyperplane, and how it contributes to the model's robustness. (4)
- C) Describe the typical neural network configuration. (2)
- 2) Criticize the three major steps involved in data analysis-data preparation, exploratory data analysis (EDA), and modelling. (4)
- A)
- B) Evaluate preprocessing steps for dominant gene prediction using ANN. (4)
- C) Illustrate how back pointer algorithm is used to minimize the error. (2)
- 3) Evaluate the suitability of Naive Bayes for classifying high-dimensional datasets (4)

- with small sample sizes by comparing its strengths and limitations to classifiers such as Support Vector Machines and Decision Trees, particularly in the context of sparse, noisy, or correlated features.
- A)
- B) Determine filter, wrapper, and embedded approaches to feature selection in a machine learning context. Demonstrate how each method performs in terms of computational efficiency, scalability, and impact on model generalization when working with high-dimensional datasets. (4)
- C) Explain how improper use of cross-validation can affect the performance evaluation of a machine learning model. What are some common mistakes and their consequences? (2)
- 4) Analyse the architecture and learning process of a Deep Belief Network (DBN), detailing the role of each layer. (4)
- A)
- B) Evaluate the effectiveness of the structured workflow used in the review of Clinical Decision Support (CDS) system papers. (4)
- C) Explain the process of identifying location of nuclei using haematoxylin stain. (2)
- 5) Evaluate the effectiveness of high-level NLP components in handling medical applications. (4)
- A)
- B) Analyse the components of an Intelligent Personal Health Record (IPHR) system and how these components interact to support personalized healthcare management. (4)
- C) Summarize the main functions and benefits of using an IPHR. (2)

-----End-----